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State Estimation Algorithms for Localization: A Survey

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Abstract: Unmanned vehicles represent a research hotspot in the fields of control and robotics. The realization of autonomous driving of unmanned vehicles requires various technologies, such as localization, mapping, path planning, and obstacle avoidance. Among these technologies, localization is a fundamental component, which can be accomplished through various methods. In this work, we focus on localization based on state estimation, as these algorithms are predominantly applied to unmanned vehicles. This paper provides a comprehensive review of state estimation algorithms commonly used for the localization of unmanned vehicles, from the perspective of control and robotic engineers. First, we provide an overview of localization schemes based on state estimation algorithms. Subsequently, we can categorize the research subjects into eight classes and clarify the principles and features of each type of state estimation algorithm. Furthermore, we examine the recent research trends associated with these algorithms.

Keywords: Finite impulse response filter, Kalman filter, localization, particle filter, state estimation.

1. INTRODUCTION

The problem of localization is aimed at determining the coordinates of an object of interest (e.g., a vehicle or a person) within a preset coordinate system through sensor measurements [1]. Specifically, two-dimensional (2D) coordinates (x, y) are estimated for objects moving on the ground, and three-dimensional (3D) coordinates (x, y, z) are determined for airborne objects. Localization may be considered similar to target tracking. However, localization is focused only on position estimation, whereas target tracking typically involves the simultaneous estimation of the position and velocity [2,3]. Furthermore, the object of interest in target tracking are typically uncooperative or hostile targets, such as preceding vehicles, enemy aircraft, and missiles. In contrast, localization is focused on friendly targets that cooperatively provide their information. Notably, the topic of interest of this research is localization and not target tracking.

Various algorithms for localization have been developed, which can be categorized into state- estimation-based and non-state-estimation-based techniques. Methods that do not rely on state estimation are mainly used in wireless communication and adopt triangulation and radio frequency finger-printing strategies based on sensor measurements to compute the position [4]. In contrast, localization based on state estimation incorporates state estimation algorithms, which were originally used in control theory to estimate the state variables of a system, for position estimation. This paper presents a review of localization algorithms based on state estimation, which will be of particular interest to control and robotics engineers.

State estimation algorithms, also known as statistical filters, represent a system as a state-space model and use noisy measurements from sensors to estimate uncertain state variables. The Kalman filter (KF) [5] is a well-known state estimation algorithm that has been successfully applied in various fields. However, the KF is designed under the assumption of a linear state-space model and Gaussian noise. Thus, it cannot be applied to nonlinear models, and its estimation performance degrades in systems with non-Gaussian noise. For localization problems in which the model for sensor measurements is nonlinear, alternative KFs have been developed, including the extended KF (EKF), unscented KF (UKF) [6], and cubature KF (CKF) [7]. Furthermore, localization systems using radio or acoustic wave-based sensors may exhibit non-Gaussian measurement noise due to the multipath effect. In such cases, the Huber KF (HKF) [8], maximum correntropy KF (MCKF) [9], and Student's t-based KF (STKF) [10] can be applied.

The particle filter (PF) is a state estimation algorithm that represents the probability distribution of a state variable as a set of random samples (particles) and implements

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the time update and the measurement update of the state variable as a process called sampling and resampling [11]. PF outperforms the KF in terms of the estimation accuracy in systems with nonlinear and non-Gaussian noise. As the number of samples increases, the accuracy of the PF increases. However, it also incurs higher computational costs which renders real-time localization more challenging than that in the case of KF. Moreover, in mobile robot localization scenario, the PF-based localization algorithm may fail or diverge if the position of the robot suddenly changes (kidnapped robot problem) or if the number of samples is insufficient relative to the space size.

Finite memory estimation (FME) algorithms are state estimation algorithms that were originally developed to address the divergence problem observed in KFs. KFs typically use all information from the initial time point to the current time point to estimate the state variable at the current time and recursively updates the state variable. Consequently, modeling or computational errors may gradually accumulate over time, resulting in filter divergence. FME algorithms mitigate this problem by estimating the state variable using only a finite number of recent measurements. Various FME algorithms, such as limited memory filters [12,13], moving horizon estimation [14], and finite impulse response (FIR) filters [15-20], have been developed, with the FIR filter garnering significant attention for the localization of mobile robots.

Overall, localization is one of the most fundamental and key technologies for autonomous driving and is expected to become increasingly important with the growing demand for unmanned vehicles. Therefore, this paper provides a comprehensive review of various state estimation algorithms for localization from the perspective of control and robotics engineers and highlights the recent research trends in the field.

The remaining paper is organized as follows: Section 2 provides an overview of localization based on state estimation algorithms as preliminary knowledge. Section 3 provides a review of the existing research on state estimation algorithms for localization, divided into eight categories to facilitate a systematic discussion. Section 4 presents the concluding remarks.

2. PRELIMINARIES: LOCALIZATION BASED ON STATE ESTIMATION

Localization based on state estimation requires state-space models that express the motion of the object and sensor measurements, referred to as the motion and measurement models, respectively. In this section, we introduce several representative motion and measurement models used in localization and describe a localization scheme based on state estimation.

2.1. Motion model

A motion model describes the evolution of the coordinates a target object over time. We introduce two simple but widely used discrete-time motion models for targets in a 2D plane. The first model pertains to wheeled mobile robots. The state vector at a discrete time k is defined as $\mathbf{x}_k = [x_k,\ y_k,\ \theta_k]^T$, where θ_k is the heading angle of the robot. The transition of the state vector can be expressed as follows [21]:

$$x_k = x_{k-1} + d_{k-1}\cos(\theta_{k-1} + \frac{1}{2}\Delta\theta_{k-1}) + w_{x,k}, \tag{1}$$

$$y_k = y_{k-1} + d_{k-1}\sin(\theta_{k-1} + \frac{1}{2}\Delta\theta_{k-1}) + w_{y,k}, \tag{2}$$

$$\theta_k = \theta_{k-1} + \Delta\theta_{k-1} + w_{\theta,k}, \tag{3}$$

where d_{k-1} and $\Delta\theta_{k-1}$ are the displacement and the increment in heading angle during the time interval $[k-1,k]$, respectively; and $w_{x,k}, w_{y,k}, w_{\theta,k}$ are the noise (or error) variables. Both d_{k-1} and $\Delta\theta_{k-1}$ can be determined by the robot control system and represent a type of control input. Because this model can be used only when the control inputs are known, it can be applied only to targets that can provide control inputs (either an ego-vehicle or a friendly vehicle). Additionally, although this model is reasonably accurate because it is based on control input information, it cannot be applied to linear state estimation algorithms owing to its nonlinear nature.

The second model is suitable for humans, who do not move in response to control inputs, or for unfriendly (or hostile) vehicles that do not provide any information regarding their motion. In the absence of any information regarding the target motion, the target is assumed to move at a constant velocity for a very short time interval T. Moreover, the target acceleration is unpredictable; therefore, the model represents it as random noise. This model, named the constant velocity (CV) motion model, defines a state vector as $\mathbf{x}_k = [x_k, y_k, \dot{x}_k, \dot{y}_k]^T$, where (x_k, y_k) and $(\dot{x}_k, \dot{y}_k)$ are the 2D positions and velocities, respectively. The transition of the state vector in the CV model [22,23] can be expressed as follows:

$$\begin{aligned}
&\mathbf{x}_{k+1} = \mathbf{F}\mathbf{x}_k + \mathbf{G}\mathbf{w}_k,\\
&\mathbf{F} = \begin{bmatrix} 1 & 0 & T & 0 \\ 0 & 1 & 0 & T \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},\\
&\mathbf{G} = \begin{bmatrix} T^2/2 & 0 \\ 0 & T^2/2 \\ T & 0 \\ 0 & T \end{bmatrix},
\end{aligned} \tag{4}$$

where $\mathbf{w}_k$ is the process noise vector, which is assumed to represent zero-mean white Gaussian noise with covariance $\mathbf{Q}_k$. The CV model is a linear model, and thus, it can

be applied to both linear and nonlinear state estimation algorithms. However, the model relies on the assumption of CV motion within a short time interval and cannot be applied in scenarios deviating from this assumption, for example, in the case of sensors with long measurement acquisition intervals or targets that exhibit sudden changes in direction. More information regarding the various types of motion models can be found in noteworthy books on target tracking [22-25].

2.2. Measurement model

A measurement model expresses the relationship between sensor measurement and state variables (positions). Among the various types of sensors and measurements that have been developed to estimate the location of a target, in this study, we introduce the simple but commonly used time-of-arrival (TOA) measurement. The TOA (or time of flight, TOF) is defined as the time required for radio waves or sound waves to travel from a transmitter to a receiver. The distance between the transmitter and receiver can be determined as the product of the speed of light or sound and the TOA. If the 2D position of the transmitter is $(\bar{x},\ \bar{y})$ and the position of the target with the receiver is represented by (x_k, y_k), then TOA measurement can be expressed as a function of the target position as follows [4,20]:

$$\begin{aligned} z_k &= \frac{1}{c}\sqrt{(\bar{x}-x_k)^2+(\bar{y}-y_k)^2}+v_k \\ &= h(\mathbf{x}_k)+v_k, \end{aligned} \tag{5}$$

where c is the speed of light (in the case of radio waves) or sound (in the case of sound waves); $h(\cdot)$ is a nonlinear function representing the relationship between the TOA and state vector $\mathbf{x}_k$; and v_k is the measurement noise. Because multiple (at least three) beacons are used for localization, multiple TOA measurements can be aggregated to form a single measurement vector. For example, using three TOA measurements from three beacons, the measurement vector can be formulated as $\mathbf{z}_k = [z_{1,k}\ z_{2,k}\ z_{3,k}]^T$. By defining a matrix function $\mathbf{h}_k(\cdot) = [h_{1,k}(\cdot)\ h_{2,k}(\cdot)\ h_{3,k}(\cdot)]^T$, the TOA measurement model can be represented as follows:

$$\mathbf{z}_k = \mathbf{h}_k(\mathbf{x}_k)+\mathbf{v}_k, \tag{6}$$

where $\mathbf{v}_k$ is the measurement noise vector, which is typically assumed to be zero-mean white Gaussian noise with covariance $\mathbf{R}_k$. Most measurement models are nonlinear. However, if radar or LiDAR systems are used to measure the relative distance and bearing angle to a target, polar coordinates can be converted to Cartesian coordinates to establish a linear measurement model.

2.3. Localization using state estimator

With the motion and measurement models defined, a state estimation algorithm can be used to estimate the position of the target. We present a simple example demonstrating position estimation using the CV motion model and TOA measurement model. The CV motion model is linear. However, the TOA measurement model is nonlinear, and thus, a nonlinear state estimation algorithm must be used. We employ the EKF, a typical nonlinear state estimation algorithm, to estimate the state vector containing 2D position information using the following equations:

$$\hat{\mathbf{x}}_k^- = \mathbf{F}\hat{\mathbf{x}}_{k-1}^+, \tag{7}$$

$$\mathbf{P}_k^- = \mathbf{F}\mathbf{P}_{k-1}^+\mathbf{F}^T + \mathbf{G}\mathbf{Q}_k\mathbf{G}^T, \tag{8}$$

$$\mathbf{K}_k = \mathbf{P}_k^-\mathbf{H}_k^T(\mathbf{H}_k\mathbf{P}_k^-\mathbf{H}_k^T + \mathbf{R}_k)^{-1}, \tag{9}$$

$$\hat{\mathbf{x}}_k^+ = \hat{\mathbf{x}}_k^- + \mathbf{K}_k(\mathbf{z}_k - \mathbf{H}_k\hat{\mathbf{x}}_k^-), \tag{10}$$

$$\mathbf{P}_k^+ = (\mathbf{I}-\mathbf{K}_k\mathbf{H}_k)\mathbf{P}_k^-, \tag{11}$$

where $\mathbf{P}_k$ is the estimation error covariance matrix; $\mathbf{K}_k$ is the Kalman gain; superscripts $-$ and $+$ denote *a priori* and *a posteriori*, respectively; and $\mathbf{H}_k$ is the Jacobian matrix computed as

$$\mathbf{H}_k = \left.\frac{\partial \mathbf{h}_k}{\partial \mathbf{x}_k}\right|_{\hat{\mathbf{x}}_k}. \tag{12}$$

3. STATE ESTIMATION ALGORITHMS FOR LOCALIZATION

3.1. Cooperative localization

Recently, cooperative localization techniques have emerged as a focus area in the field of multiple-robot localization. In cooperative localization, multiple mobile robots collaborate with one another to estimate their positions. Here, cooperation primarily involves the sharing of information among the robots. Equipping each robot with expensive and precise sensors for collaborative missions may be cost-prohibitive. The operating cost can be reduced by equipping only a few robots with high-precision sensors, while outfitting the remaining robots with low-cost sensors that can gather relative measurements to estimate their position through collaborative efforts. In such scenarios, the robots with high-precision sensors are referred to as leader robots, which can independently perform localization without the aid of other robots. Non-leader robots possess low-cost sensors that can measure only the relative distance and bearing angle of their neighboring robots. With this relative measurement information, non-leader robots can infer the coordinates of other robots relative to their own position, equivalent to estimating the position of a target in a target tracking problem. If a neighboring robot knows its absolute coordinates and shares this information with a non-leader robot, the non-leader robot can determine its position from the position of the neighboring robot and relative measurements. To this end, some robots (i.e., the leader robots) must be able to obtain their absolute positions without the assistance of other robots and share this information with robots

through communication, allowing other robots to systematically estimate their positions.

The abovementioned strategy represents a typical approach for cooperative localization. In the following text, we introduce state estimation algorithms for cooperative localization. In cooperative localization, individual robots use sensors to acquire absolute and relative measurements and communicate this information with their neighbors. Based on the collected information, each robot individually estimates its own position. The state estimation algorithm commonly used in such localization scenarios is distributed filtering. Notably, a distributed KF algorithm that integrates absolute and relative measurements was proposed for cooperative localization in [26]. Furthermore, a distributed FME (DFME) algorithm that leverages relative measurements was introduced in [17], using the average consensus algorithm to process the measurement data. In general, the occurrence of communication failures during the transmission of measurement information from a robot to other robots can degrade the localization accuracy. To address this problem, a DFME algorithm was proposed [18]. In this framework, a measurement model reflecting the missing measurement was designed, and the gain matrices for the DFME were derived using the measurement model. Additionally, if the control system that governs multiple robots is positioned in a remote location, it is necessary to transfer information from the robots to the control system. Two key issues may arise in this information transfer process: First, the transmission of a large amount of information may result in delays or data loss. To mitigate the communication burden, event-triggered mechanisms can be used, which allow transmission only if certain conditions are satisfied. In [27-29], various event-triggered localization algorithms based on the EKF were explored. Second, a hostile group may launch cyber-attacks, which can disrupt the delivery of information or manipulate its content. To address this security concern [30,31], cooperative localization algorithms based on EKF that are robust against cyber-attacks have been developed.

3.2. Asynchronous localization

The signal transmission time between stationary anchor nodes (beacons) and mobile nodes can be used for localization based on sensor networks, with representative methods being TOA and time-difference of arrival (TDOA)-based localization schemes. Accurate measurement of the transmission time necessitates the synchronization of clocks of the components constituting the anchor and mobile nodes. Because radio waves cannot propagate underwater, sensor-network-based localization systems rely on sound waves. However, complex environmental factors may introduce propagation delays, with an effect similar to that of asynchronous clocks, which can adversely affect the transmission of sound waves underwater. To overcome this problem, a localization algorithm based on the consensus strategy and UKF was proposed in [32]. The authors examined the relationship between the propagation delay and position and attempted to compensate for the effect of asynchronous clocks through a careful design of the consensus-based UKF. In [33], the problem of localization using unsynchronized beacons was addressed. Unlike [32], which presented an algorithm robust to the effects of asynchronous clocks, the authors in [33] proposed an algorithm that estimates the time offset between transmitters and receivers. A local optimization algorithm was used for the self-calibration of the time offset, and a combination of KFs and UKF was used for localization.

3.3. Reducing computational burden in localization

A notable disadvantage of PFs compared with KFs is their high computational cost. As the number of samples (i.e., particles) increases in PFs, the estimation accuracy increases, albeit at the expense of increased calculation. Therefore, several researchers have attempted to minimize the number of samples while maintaining a satisfactory level of estimation performance. Adaptive PFs have been widely used for enhancing the computational efficiency by adjusting the number of samples according to the application. In [34], the Kullback-Leibler distance (KLD)-sampling method was used to adjust the sample size. This approach approximates the probability distribution using samples in a PF, measures the approximation error using the KLD, and adjusts the number of samples to maintain the error within a certain value. When the KLD is high, more samples are considered to reduce the error, and if the KLD is low, fewer samples are used to maintain the accuracy. PF-based localization algorithms are typically referred to as Monte Carlo localization (MCL), and various MCL algorithms based on KLD sampling have been explored [34-36].

Another type of adaptive PF is the self-adaptive MCL (SAMCL) [37,38]. SAMCL focuses on the spatial distribution of samples. Because the location of the target is uncertain in the initial stages of localization, the samples are widely scattered across the space. As the location of the target becomes more certain, the samples cluster around the position of the target. If the location of the target is uncertain and the samples are not widely dispersed, the target may be lost. Consequently, many samples are required to cover a large area. SAMCL reduces the number of unnecessary samples by preventing them from being spread over an unnecessarily large area and concentrating them where the target is most likely to be found.

Box particle filters [39,40] represent a different approach for reducing the computational cost of PFs. PFs typically employ a specific value of the state vector as a particle, i.e., a point particle. In contrast, box particle filters use intervals of values that the state vector can take.

For example, for a 2D and 3D state vector, the interval can be represented by a rectangle and a box, respectively. A box particle contains an infinite number of point particles. Box particle filters can effectively address non-Gaussian or biased noise and can help reduce the number of particles required for accurate localization.

3.4. Improving localization accuracy

Received signal strength indicator (RSSI)-based localization uses the measured signal strength values to estimate the location of a target [41]. Localization methods based on RSSI can be divided into fingerprinting and distance-based methods. The fingerprinting approach involves storing predetermined signal strength values corresponding to known locations and comparing them to the received values [42]. The distance-based method uses signal attenuation models and signal strength measurements to compute the distance between the transmitter and receiver, thereby enabling location estimation based on this distance information. The system for RSSI-based localization is simple and inexpensive to implement because it does not require clock synchronization between sensors and does not transmit any additional information within the signal. However, it is typically less accurate than other methods, such as TOA and TDOA based schemes. Localization algorithms based on smoothing have been studied to address these limitations and develop a low-cost, high-accuracy localization system.

In the context of localization, filtering involves using measurements up to time instant k to estimate the state variable at that time, whereas smoothing involves the use of measurements up to time $k+N$ to estimate the state variable at time k. Smoothing is more accurate than filtering because the state variables are estimated by leveraging more information. Three types of smoothing methods are commonly used: fixed-point, fixed-lag, and fixed-interval smoothing. Fixed interval smoothing uses all measurements within a specific interval to estimate the state variable at each point in time within that period. One widely used fixed interval smoothing approach is Forward-backward smoothing, which uses two types of filters. The forward filter estimates the state by following the time sequence, whereas the backward filter corrects errors retrospectively using an inverse dynamic model. Subsequently, the smoothing method combines the estimates from both types of filters to yield the final state estimate. In [43], a sigma point Kalman smoother tailored for RSSI-based indoor localization systems was proposed. Visible light positioning (VLP) systems are RSSI-based localization systems that use the intensity of light from a light source. Because conventional ceiling lights can be used as markers, VLP systems can be constructed in a cost-effective manner. However, it is challenging to precisely model the light attenuation in complicated indoor environments, and thus, additional algorithms must be used to increase the localization accuracy. In [44], a Kalman smoother for VLP systems was proposed, using an ensemble KF (EnKF) and information KF (IKF) as forward and backward filters, respectively. Smoothing techniques may also be based on PFs. In [45], a Monte Carlo smoothing approach for indoor localization was proposed. Additionally, in [16], an FME-based smoothing technique was developed to improve the localization accuracy of a quadrotor unmanned aerial vehicle. In this algorithm, an extended unbiased FIR filter was used as a forward filter for robust localization, and an unbiased FIR smoother was used as a backward filter to improve the accuracy.

3.5. Recovery from localization failure

In applications involving small mobile robots, such as robot soccer, a phenomenon known as robot kidnapping may be encountered, in which the robot position artificially shifts to a remote location. In such events, KF-based localization algorithms temporarily exhibit a dramatic rise in the positioning error, which subsequently reduces and returns to normal over time. However, in PF-based MCL algorithms, robot kidnapping frequently results in localization failure and divergence of the positioning error. In particular, PFs use samples to represent the probability distribution of a state variable. In the initial stage of localization, the samples are widely distributed in space, but the samples converge around the actual target (robot) location over time. In robot kidnapping scenarios, if the robot location changes to a remote area with few samples, the samples near the former location of the robot are not regenerated during the resampling process and disappear. The lack of adequate samples near the new location of the robot and disappearance of samples in the former site may lead to complete localization failure. This problem, which is commonly encountered in MCL algorithms, is termed the *kidnapped robot problem*, and various strategies have been developed to address it.

One such strategy is the mixture MCL method [46], which reverses the order of the sampling and resampling processes of the PF. In standard MCL algorithms, first, sampling is performed to accomplish the time update of samples using the motion model, and then, resampling is conducted to reselect the samples with a probability proportionate to the likelihood based on the measurement data. In the case of robot kidnapping, the measurements are obtained based on the location to which the robot moved, and thus, the samples at the original location of the robot have low likelihood and are lost in the resampling process. The mixture MCL algorithm implements dual MCL by first generating samples around the actual robot location based on the measured values and then updating them using the motion model. The results of dual MCL are combined with those of regular MCL to improve the robustness of the localization algorithm. Even if the regular MCL fails due to the robot kidnapping phe-

nomenon, dual MCL can still recover. In normal conditions without robot kidnapping, the mixture MCL can still leverage the more accurate localization results offered by regular MCL.

The reverse MCL framework [47] is similar to the mixed MCL in that it reverses the order of sampling and resampling processes. However, it implements the reverse algorithm in a different manner. Specifically, reverse MCL uses a basic but less accurate localization algorithm known as 2D Markov localization to identify likely locations of the robot and then implements regular MCL based on this information. This approach not only improves the localization accuracy but also effectively addresses the kidnapped robot problem.

Hybrid filters and fusion filters combine two complementary types of filters. Specifically, hybrid particle/FIR filters [19] combine particle and FIR filters to achieve accuracy and robustness. The model used to represent the sensor measurements for localization is frequently nonlinear and contains non-Gaussian noise. PFs are well-suited state estimation algorithm for such models. However, as discussed previously, PFs may undergo complete failure in the case of robot kidnapping or in scenarios involving sample impoverishment. This problem can be solved by combining a PF with a less accurate but more robust filter. FIR filters exhibit limited filter divergence because they use only recent finite measurements. Moreover, these filters demonstrate global localization capabilities that allow localization to be performed in batches using only recent measurements without the need to set and recursively update an initial estimate. However, FIR filters are less accurate than PFs. In a hybrid particle/FIR filter algorithm, a PF with high accuracy is used as the primary filter, and a FIR filter with high robustness is used as the secondary filter. In normal conditions, only the PF is used for localization. However, if the localization accuracy of the PF deviates significantly, the FIR filter is activated. PF failure is detected using the Mahalanobis distance between the predicted and actual measurements. A scenario is considered to correspond to abnormal failure if this distance exceeds a predetermined threshold, which is typically determined using the chi-square distribution table. When the PF fails, the FIR filter is reset and reactivated by generating a random sample based on the state estimate and estimated error covariance value obtained after the FIR filter is run.

3.6. Localization in NLOS situations

Indoor localization systems rely on radio waves, sound waves, and light signals to obtain the measurements required for position estimation. However, NLOS conditions may occur due to obstacles between the transmitter and receiver that can interfere with signal transmission. In NLOS situations, the signal transmission may be completely blocked, and no signal may be received. In such cases, localization must be performed in the absence of some or all measurements. Furthermore, radio waves or sound waves may be transmitted via indirect paths through reflections off walls, ceilings, or objects. In this case, the signal may arrive at the receiver later than expected, resulting in time delays and bias errors in the TOF-based distance measurements. NLOS issues can be addressed using two approaches: NLOS detection and mitigation. NLOS detection is aimed at identifying and managing NLOS situations or NLOS-biased measurements. Mitigation involves the use of a position estimation algorithm that is robust against NLOS effects. NLOS conditions can be easily detected by setting a threshold and identifying TOF measurements that are abnormally increased (with a bias error) compared with measurements from previous time points. More sophisticated methods involve intelligence-based algorithms, such as pattern classification. Because this work focuses on state estimation algorithms, we do not discuss NLOS detection and instead review position estimation algorithms that are robust against NLOS effects. Indoor localization systems typically receive signals from multiple anchor node (beacons) and use multiple measurements to estimate the target position. When some measurements are found to be contaminated by NLOS (i.e., they contain bias errors), it is desirable to use only those measurements to establish a region where the target can be located. Then, constrained filtering can be performed using the remaining uncontaminated measurements [48].

To mitigate NLOS effects, a distributed localization algorithm based on a hybrid particle/FIR filter was proposed [20]. This algorithm detects and avoids NLOS measurements using distributed filtering and data association techniques, and it leverages the ability of the hybrid particle/FIR filter to recover from failures when NLOS effects cause a sharp increase in the localization error.

Instead of detecting NLOS situations, some researchers assumed that they are unpredictable and developed position estimation algorithms that are robust against NLOS effects. For example, in [49,50], robust filter-based localization algorithms were designed for mixed line-of-sight (LOS)/NLOS situations. In [49], a measurement model for mixed LOS/NLOS situations was established and used to design robust filters. In [50], the transition between LOS and NLOS situations was represented using a jump Markov model.

Other localization algorithms [17,18,51,52] have also been proposed to cope with time delays or lost measurements due to NLOS. In [17,18], the lost measurements were predicted based on the estimated state at the previous time step. Furthermore, a measurement model that simultaneously considers both the predicted and actual measurements was designed, and a FIR filter was derived based on this model. Radar or LiDAR sensors can also be used to obtain the distance measurements required for localization. Notably, a radar system may receive reflected

radio waves from non-target objects, which are known as false measurements. Signals below the detection threshold can be filtered out to eliminate the false measurements. However, if the threshold is set too high, all the measurements may be filtered out. In [52], measurements were modeled using a random finite set (RFS), and a PF was designed based on the RFS to manage false/missed measurements.

NLOS situations may also occur in outdoor localization using the global positioning system. For example, NLOS may result in time delays or measurement loss when a vehicle equipped with a navigation system traverses a road surrounded by high mountains or buildings. To address this problem, a novel variational Bayesian (VB)-based KF was proposed in [51]. Two types of Bernoulli variables were used to determine the time delay and measurement loss. Furthermore, the VB technique was applied to simultaneously estimate both the delay time and state variables.

3.7. Overcoming uncertain measurement noise

Most of the commonly used state estimation algorithms rely on Gaussian noise models. However, the measurement noise in real-world localization systems does not always follow a Gaussian distribution. For example, if radio waves or sound waves are reflected or refracted instead of traveling in a straight line, the measurement noise may deviate from a Gaussian distribution. This situation is especially evident when signals are transmitted over acoustic waves underwater: The complex interplay of factors such as salinity and water temperature causes sound waves to reflect and refract. If the actual measurement noise is non-Gaussian, the accuracy of state estimation algorithms designed under the assumption of Gaussian noise may be degraded. Various state estimation algorithms have been explored to address the uncertainty in measurement noise.

For example, in [53-56], position estimation algorithms for autonomous underwater vehicles (AUVs) were developed for a cooperative localization system in which a surface ship serves as the leader and collaborates with multiple AUVs. To handle uncertainties in the measurement noise covariance, an adaptive EKF was proposed that estimates the covariance using an expectation maximization algorithm [53]. In [54], measurement noise was modeled as a heavy-tailed mixture distribution, whereas in [55], it was modeled as a Gaussian-Slash mixture distribution and Gaussian generalized hyperbolic skew Student's t mixture distribution. VB inference was used to derive robust KFs based on the non-Gaussian measurement noise models [54,55].

Typical state estimation algorithms, such as the KF, PF, and FIR filters, are unbiased estimators, which means that the expected value of the estimation error is zero. Unbiased estimators can realize accurate localization if the error is appropriately modeled. However, if the noise is inaccurately modeled, the accuracy of the unbiased estimator may be degraded.

Notably, biased estimators, such as trilateration frameworks [57], min-max algorithm [58], centroid method [59], and geo-n method [60], are less accurate than unbiased estimators, but they are not affected by the noise model and are less computationally intensive. In [61], a biased constrained hybrid KF (BC-HKF) was proposed for range-based localization. In this approach, the initial position estimate is obtained using the min-max algorithm, which is a biased estimator, and the constraint region is derived based on this estimate in the BC-HKF algorithm. This strategy converts the nonlinear estimation problem into a linear estimation problem, and the position is re-estimated using the KF to reduce the bias error. Finally, a sigma point within this region is chosen to obtain the final position estimate using the UKF.

3.8. Simultaneous localization and calibration

Addressing the model uncertainty is a critical concern in the context of localization. One particular challenge pertains to the measurement noise models, as discussed previously. Simultaneous localization and calibration (SLAC) algorithms are localization techniques that estimate unknown (potentially time-varying) model parameters and calibrate them in real time. The relationship between sensor readings and a state variable (position) is modeled for localization. However, real-world measurements may deviate from this model owing to several factors. For example, in underwater localization based on sonar sensors, the refraction and reflection of sound waves caused by salinity and water temperature may introduce errors in the sound wave attenuation model. In the case of mobile robots, differences between the control commands and actual rotation speed of the robot wheels may lead to errors in odometry. In radio-wave-based localization systems, time delays can occur due to multipath effects, which are difficult to predict because they depend on the environment. To reduce these errors, sensors must be assessed and calibrated in advance. These tasks are typically performed once before localization. However, even pre-calibrated sensors may be vulnerable to errors owing to environmental changes. SLAC algorithms, which can simultaneously perform position estimation and calibration in real time, have emerged as promising tools to overcome this problem.

In typical SLAC frameworks, the parameters that need to be calibrated are added to the state vector, thereby generating an augmented state vector and a state-space model based on it. Subsequently, a state estimation algorithm is used to simultaneously estimate the position and unknown calibration parameters. However, the state-space model has high dimensionality and nonlinearity. PFs are well-suited for highly nonlinear state-space models. However, they cannot effectively estimate state vectors with high dimensionality. In [62], SLAC was implemented using

the Rao-Blackwellized PF (RBPF), which is suitable for high-dimensional state estimation. The bias of the global navigation satellite system (GNSS) and vehicle position were jointly estimated using RBPF. In [63], the antenna response and ranging bias of the GNSS were incorporated in the state vector for calibration, and an iterative EKF was used to manage the model nonlinearity. Another challenge arises during the simultaneous estimation of calibration and location variables through state vector augmentation: The observability for all variables in the augmented state vector is not guaranteed. To solve this problem, the decoupled estimation technique was proposed [64], which separately estimates the variables related to localization and calibration. Specifically, a linear constrained state estimator was used to estimate the state variables under parameter uncertainty, and the uncertain parameters were estimated using a recursive least square estimator. In a distance-based indoor localization system, the coordinates of beacons, termed access points (APs), must be known in advance. Because ultrasonic beacons are inexpensive, numerous beacons can be installed to increase the accuracy of location estimation. However, it becomes challenging to accurately measure the absolute coordinates of all the installed beacons. To address this problem, some researchers recommended the accurate calibration of APs for localization and estimation of the absolute coordinates of uncalibrated APs based on this information [65].

4. CONCLUSION

This paper provides a comprehensive review of state estimation algorithms applied in localization applications. By analyzing the reviewed literature, we identified the following trends in this research domain. First, traditional research has been focused on enhancing the localization accuracy and reducing the amount of computation to enable real-time processing. Second, considerable attention has been directed to algorithms that combine filters with different properties to achieve a balance between accuracy and robustness against model uncertainty in localization application. Lastly, recent research has been actively focused on distributed filtering using relative measurements for cooperative localization, event-triggered mechanisms to reduce communication burden in remote localization, algorithms for cyber-attack resistance, and high-dimensional and highly nonlinear filtering techniques for SLAC.

CONFLICT OF INTEREST

The authors declare that there is no competing financial interest or personal relationship that could have appeared to influence the work reported in this paper.

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